Rank-27 Vivify Cross-Band Reclamation Repair

Tactix Development Project

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Executive summary

Source-backed tracing localized a portable Vivify throughput gap to physical arena reclamation after each tier band. Kissat keeps shortened records in place across the dose and reclaims once afterward; Tactix performed a logical-order collection after every changed band. The accepted repair keeps one rule-owned reclamation transaction across the DefaultPolicy dose, rebuilds the next band's watches in the same logical order when records are still in place, and performs one collection at dose end. Canonical public Vivify still creates and drains its own transaction, preserving the complete N1 action surface and the proof-rule/strategy boundary.

The exact source diff has SHA-256 `a8cd5650a880948cafa928120bf7d0eb36cc103aa081626a2c9b07977ee35fdd`; the clean Release binary has SHA-256 `4b3f7467b69e165686e271124d79a6b4ba977c2d6829a72afacf19ec23bc5e6a`. An exact Rank-27 prefix preserves the 64-row event trace and lowers Vivify from 1.730 to 1.146 seconds. Independent full Rank 27 improves wall time 433.41 to 427.29 seconds and Vivify 52.515 to 38.473 seconds. The complete B28+C5 gate passes 66/66 independently checked tasks at 0.968834x predecessor PAR-2 with zero solve loss, invalidity, cleanup failure, or certificate residue. Claude Code's authoritative exact-source fuzz verdict is terminal PASS: 30,000 generated cases and 81 corpus replays produced no sanitizer, crash, invariant, model, proof, or differential failure.

The Release suite is 73/74. Every functional, proof/model, trace, boundary, reset, cleanup, and hygiene test passes; only the already recorded opt-in documentation-size policy debt fails. This closes the isolated Vivify increment. It does not close convergence or make 300 seconds a stopping criterion.

1 Owner and implementation

The accepted behavior before this increment already gave Vivify stable radix ordering and exact candidate-order parity. Traces of the remaining broad throughput gap showed repeated reclamation between independently budgeted tier bands. Tactix's proof rule appends a packed replacement and historically collected immediately to recover logical clause order. Kissat's corresponding dose retains shortened records until all bands have run.

The repair introduces `VivifyReclamationBatch`, whose only state records whether retirement or inheritance requires later collection. Every band runs through the ordinary registry dispatch frame using `apply_rule_with`; the alternate entry point changes no guard, timer, trace, journal, or proof-rule framing. If an earlier band changed records, the next band rebuilds large watches from persistent logical source-position keys, reproducing the watch stream that per-band logical-order collection would have produced. The policy then collects once after the dose unless the solver is inconsistent. Direct `vivify_apply` and `vivify_try` own and drain a local transaction, so standalone rule semantics remain complete.

The change contains no benchmark name, identity, rank, time limit, per-instance arm, or tuned constant. It changes storage timing only after the exact candidate/effect transaction has established the same logical order. Proof addition/deletion, reconstruction, scheduling budgets, candidate order, and N1–N4 remain unchanged.

2 Exact state and long-horizon evidence

Exact Rank-27 prefix jobs `1647788/1647790` preserve the accepted 64-row event trace byte-for-byte at SHA-256 `7a6216bb86c236512dea03de76c4c9ab2ffdd745bd7ca7b9fa5507e708987cc1`. Candidate order, tick and cost ledgers, scheduling order, learning, pruning, and resulting state are exact over the common prefix. Vivify time falls from 1.730 to 1.146 seconds.

The six-instance mechanism screen `1647791/1647798` passes 12/12: three SAT models and three UNSAT proofs validate independently per arm, Vivify time falls on all six witnesses, candidate/predecessor PAR-2 is 769.208/770.504 seconds (0.998318x), and cleanup and certificate residue are zero. Some complete trajectories are nonexact, so this screen is a long-horizon guard rather than an exact-state attribution claim.

Independent full Rank-27 jobs `1647806/1647807` each use two CPUs and return valid UNSAT with zero residue. Candidate/predecessor wall time is 427.29/433.41 seconds, internal time 426.802/432.912, Vivify 38.473/52.515, inprocessing 142.955/151.328, and peak RSS 606,616/688,076 KiB. Printed effects remain identical except final search-mode ticks, which differ by 12 out of about 10.35 billion. Follow-up source-aligned tracing attributes the first favorable and later unfavorable tick crossings to one fewer large-record relocation and the bounded cost of rebuilding watches over retained physical records. The full run is therefore positive performance evidence, not an exact-complete-state proof. Exact prefix identity and the complete correctness gates carry the semantic claim.

3 Consolidation and correctness gates

Frozen campaign `1647857/1647866` runs the complete B28+C5 matrix as 66 independent one-solver array tasks under the two-CPU, 16-GiB, 1200-second solver contract. All 66 results are valid. Each arm independently checks 17 SAT models and verifies 16 UNSAT proofs against the original input. Candidate/predecessor PAR-2 is 1477.077/1524.592 seconds, ratio 0.968834; maximum RSS is 2.930233/2.930515 GiB. There is no missing or malformed task, unexplained solve loss, invalid model or proof, cleanup failure, or certificate residue. The frozen summary has SHA-256 `4ef52978711634f0470c547ed96e397b42716e0819c877f1c12785a418d19925`.

Claude Code exclusively prepared, submitted, monitored, inspected, and classified exact-source fuzz campaign `20260826-0103-vivify-crossband-reclamation-fuzz`. Its authoritative terminal verdict is PASS. Prep `1647924` rebuilt an ASan/UBSan Debug pin at SHA-256 `93997df953a33eabcdf256baf3a8ffab1257e8d1c497bd8df1b46697e550ecc4` and reverified source identity. Independent legs `1647925`–`1647928` complete 81/81 corpus replays and 30,000/30,000 generated cases. Of 26,686 oracle-bearing verdicts, 15,797 SAT models validate, 10,889 native LRAT proofs verify, and pinned Kissat agrees on every verdict. The remaining 3,314 cases are declared bounded five-second semantic nonterminations and make no oracle claim. Default, stress, and random legs all reach Vivify; eight direct traces hit both changed cross-band paths. Claude reports zero sanitizer finding, crash, invariant abort, rejected model or proof, differential disagreement, certificate residue, or job residue. Codex accepts that report without probing fuzz artifacts.

4 Rejected consolidation investigations

The isolated Substitute call-level deferred-reclamation prototype preserves a short exact trace but adds 1,800 ticks and regresses full Rank-27 internal time 552.836 to 562.824 seconds while increasing memory. It is rejected and fully reverted. Residual BVE is not presently a demonstrated common-state worse-side owner: refreshed current Tactix spends 15.958 seconds versus current Kissat's 16.32 seconds, while their trajectories differ; the last exact BVE weakness, antecedent admission, is already repaired. No BVE source change is retained.

Residual Congruence remains a measured but unowned gap: the refreshed matched prefix is about 1.830 seconds versus current Kissat's 1.644 seconds. Three- inline and two-inline closure layouts preserve exact prefix state but do not improve full Rank 27; the latter regresses wall time 544.52 to 546.69 seconds and raises observed RSS. Both are rejected and fully reverted. Generic key- layout guesses are closed; any return must first localize exact closure matching or maintenance work.

5 Disposition and next work

The terminal gate and fuzz PASS promote cross-band Vivify reclamation as a portable common-state representation repair. The next convergence increment returns to broad search quality against the accepted current-Kissat comparator: freeze this binary as the new Tactix baseline, identify the earliest exact common-state point where equal cost yields weaker learning or pruning (or equal logical work costs more), and trace decision, propagation, restart/reduce, and inprocessing schedule order before changing behavior. Residual Substitute maintenance and the smaller Congruence closure gap remain documented owners only when exact tracing demonstrates them. Every future repair remains isolated through exact prefix, full Rank 27 plus mechanism-relevant B+C, the complete checked gate, Claude-owned fuzz, and a separate report/commit.